CHAPTER 7
Operational Frameworks

7.1 CONTROL OF THE MONEY SUPPLY

The short-term liabilities of central banks, whether it is bank notes issued by them, or the accounts held by commercial banks at the central bank, form the basis of money, also known as base money in most developed economies. Central banks have some, but not unlimited, control over these liabilities. One can classify actions of central banks into three groups according to the effect they have on their total liabilities. Some actions have no effect, for instance the exchange of current account holdings at the central bank into bank notes. Some actions increase central bank short-term liabilities, such as the purchase of assets by the central bank. These are known as liquidity providing operations. The third group are actions that reduce short-term liabilities, such as asset sales, known as liquidity absorbing operations.

Each central bank has a set of pre-defined operations that fall into each of these categories and together form the operational framework of that central bank. Because they are predefined, market participants, notably commercial banks, can form reasonable expectations about the conduct of these operations and the risk of unintended side effects is minimised. While the operational frameworks of central banks tend to exhibit strong hysteresis, some desirable common features can be identified. A good framework should at the same time be wide enough to cover potential emergencies without the need to introduce ad hoc tools, and parsimonious enough to do so with the minimum of complexity [10]. As a result, operational frameworks can be broadly divided into three parts. The first are standard operations which are intended to be used on a regular basis and the parameters of which signal the general monetary policy stance in normal times. A second set of operations is used in emergency situations as part of the LOLR function of the central bank. The third set are non-standard operations that are used broadly, but only at times of particular economic exigency. Each central bank can assemble these three sets of operations in different ways. For instance, a standing lending facility may be part of the standard operations for one central bank, while it forms part of the LOLR framework of another.

The following two sections describe some standard monetary operations tools with no claim to completeness.

7.2 LIQUIDITY PROVISION: REDISCOUNTING, OUTRIGHT PURCHASES AND LOMBARD LENDING

Central banks can in principle provide liquidity to commercial banks by lending to, or buying assets from, them. When central banks lend to commercial banks, they generally do so against collateral[1].

Central banks provide liquidity by discounting the commercial bills presented by banks, i.e., by purchasing these bills for cash. This is known as rediscounting. This term is based on the idea that the commercial bank in question first obtained the bill by discounting it. It should be borne in mind that the legal mechanics of bill trading imply that the bank presenting the bill for rediscounting can be subject to claims for redress from the central bank if the draftee of the bill refuses payment. The central bank therefore has recourse to both the bank it advances cash to, through rediscounting, and to the ultimate borrower. The practice of rediscounting bills has become less prevalent in modern-day central banking but explains terms such as the discount window at the United States Federal Reserve.

The rediscounting of a bill provides liquidity to the general economy for as long as the bill is outstanding because the payment of the bill to the central bank will amount to a reduction of the asset and liability sides of the central bank. Given the short-term nature of bills, the central bank tends to be passive in the repayment profile of its assets and would have to offset this passivity through an active purchase strategy to target a given liquidity supply.

Central banks can choose instead to purchase long-term assets from the market and resell them before maturity to gain better control over the total amount of liquidity in the banking system. The buying and selling of securities in the open market is known as open market operations or outright operations. This is the standard operating mode of the Fed.

In another type of liquidity provision, the central bank advances cash against securities posted for the duration of the lending operation. This was known as Lombard lending but today the term is somewhat archaic. In modern financial markets, the equivalent transaction is a sale combined with a simultaneous repurchase of the same security for future settlement. This type of trade is known as repurchase or simply repo trading. Where central banks lend against collateral, rather than purchasing it outright, they nowadays do so in line with repo market practice. The ECB and Bank of Japan operate in this manner in normal circumstances.

Rediscounting and open market operations are similar in the sense that the central bank becomes outright owner of an asset in exchange for liquidity injected into the banking system. They differ in that the instrument purchased is usually held to maturity in rediscounting operations while it may be sold in open market operations. Open market operations are similar to repo operations in that the maturity of the operation is shorter than that of the instrument used, but they differ because the maturity of a

[1] Lending without any form of security would create unacceptable moral hazard. Banks could advance loans to related parties and then cover any losses through borrowing from the central bank and ultimately socialise any losses.

repo operation is usually fixed while the re-sale of an asset purchased in an open market operation is at the discretion of the central bank. Open market operations, unlike repos, expose the central bank to price risk.

Although the recent large-scale asset purchase programmes have blurred the lines somewhat, it can be said that the US Federal Reserve relies mostly on open market operations while other central banks use repos for their liquidity providing operations.

7.3 LIQUIDITY ABSORPTION: ASSET SALES AND REVERSE REPOS

Just as asset purchases by central banks supply liquidity to the economy, asset sales by the central bank pull liquidity back from the economy. Central banks can sell securities issued by other entities provided that they have previously been purchased by the central bank, or they can issue securities themselves and sell those. As with asset purchases as a liquidity absoprtion tool, the maturity of the securities involved creates a predefined maturity patterns to the liquidity effect that can be counteracted by active portfolio management by the central bank.

Repo operations, in contrast, decouple the maturity of the securities involved from the maturity of the liquidity impact. This decoupling can only be taken so far given that the repo markets are liquid only up to horizons of a few months.

7.4 THE IMPACT OF FX OPERATIONS

Central banks, or other national authorities tasked with managing the external value of the currency, routinely transact in the foreign exchange market. When they buy foreign currency from domestic entities in order to manage their own currency downwards, they create central bank liabilities. Sales of foreign currency conversely reduce central bank liabilities. Interventions in the FX markets therefore interact with the purely domestic factors affecting liquidity in the domestic money market. Such operations can clash with the primary mandate of the central bank in question, such as inflation targeting.

This problem can be solved to some extent by sterilisation of FX interventions. This type of operation reduces liquidity by exchanging a liquid asset for a less liquid asset. For example, the Ministry of Finance of Japan, which is responsible for intervening in the FX market, has accumulated large amounts of foreign currency. In order to buy this foreign currency, it has to pay Japanese Yen. Because the Ministry of Finance is not the central bank, it obtains the required Yen amounts by issuing so-called Foreign Exchange Fund Financing Bills. Selling these bills on the open market turns cash into less liquid bills. The selling of Yen against foreign currency therefore does not increase the amount of Yen cash in the market because it is offset by the issuance of financing bills.

For a central bank that issues domestic currency for FX intervention purposes, the more direct way to sterilise the impact on the domestic money supply is to buy domestic long-term assets. Again, while in the first instance liquidity is created for the purpose of buying foreign currency, it is in a second step transformed into less liquid assets on the balance sheet of the private sector. This type of transaction is a large contributor to the so-called NMP (non-monetary policy) portfolios of some euro area central banks.

FX interventions are symmetric in the sense that they involve both currencies in a currency pair. A central bank that sells, for instance, euros to buy dollars on the open market increases the supply of euro liquidity in the private sector while reducing the supply of dollars. If the central bank in question were the ECB, the sterilisation on the increased euro liquidity would be in the interest of the ECB, and presumably managed by it, because that liquidity impact might affect its inflation mandate. The impact on the dollar market, meanwhile, would have to be managed by the US Federal Reserve. As a result, FX interventions among the major currency areas are usually coordinated in order to contain the spillover effects on the money markets concerned.

A case in point is Japan, which has a persistently positive current account. This means that Japanese entities tend to earn more money abroad than they have to spend domestically. This leads to an interconnection between the overseas and domestic money markets. Generally, interest rates in Japan are depressed by a generous availability of yen funding versus a variable supply of dollar funding. It can be argued [80] that this restricts the ability of the Bank of Japan to control effective short-term interest rates in its own currency. The interest rates on short-dated Japanese government debt turned negative even before the Bank of Japan cut its own policy rates below zero.